



Summer 2024

Exam Date: 28.07.2024

Exam Time 60 min.

MEP221s: Fluid Mechanics and Turbomachinery
Midterm Examination

The Exam Consists of **three** Questions on **Three** Pages

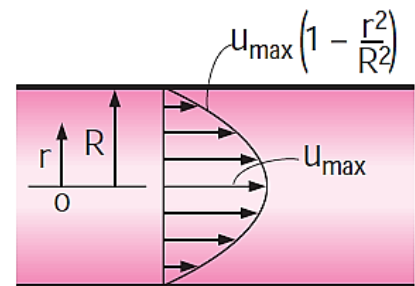
Total Marks: 25 Points

Student Name:

ID:

Question (1): (7 points)

In the region far from the entrance, fluid flow through a circular pipe is one-dimensional, and the velocity profile for laminar flow is given by $u_{(r)} = u_{\max} (1 - r^2/R^2)$, where R is the radius of the pipe, r is the radial distance from the center of the pipe, and $u_{\max}$ is the maximum flow velocity, which occurs at the center. Determine the drag force for water flow with $R = 0.08$ m, $L = 15$ m, $u_{\max} = 3$ m/s, and $\mu = 0.0010$ Pa.s.

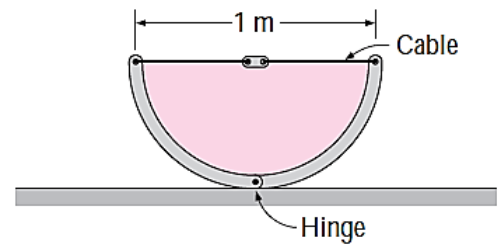


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Question (2): (9 points)

A water trough of a semicircular cross-section of radius 0.5 m consists of two symmetric parts hinged to each other at the bottom. The two parts are held together by a cable and turnbuckle placed every 3 m along the length of the trough. Calculate the tension in each cable when the trough is filled to the rim.

(Note: The centroid of a quarter-circle is $4R/3\pi$ away from the centerline)



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Question (3): (9 points)

A 1-m-diameter, 2-m-high vertical cylinder is completely filled with gasoline whose density is 740 kg/m^3 . The tank is now rotated about its vertical axis at a rate of 90 rpm while being accelerated upward at 5 m/s^2 . Determine:

- The difference between the pressures at the centers of the bottom and top surfaces
- The difference between the pressures at the center and the edge of the bottom surface.

